

4.12.3 Lists in functional programming

4.12.3.1 List processing

Content	Additional information
Be familiar with representing a list as a concatenation of a head and a tail. Know that the head is an element of a list and the tail is a list. Know that a list can be empty. Describe and apply the following operations: • return head of list • return tail of list • test for empty list • return length of list • construct an empty list • prepend an item to a list • append an item to a list. Have experience writing programs for the list operations mentioned above in a functional programming language or in a language with support for the functional paradigm.	For example, in Haskell the list [4, 3, 5] can be written in the form <i>head:tail</i> where <i>head</i> is the first item in the list and <i>tail</i> is the remainder of the list. In the example, we have 4:[3, 5]. We call 4 the <i>head</i> of the list and [3, 5] the <i>tail</i>. [] is the <i>empty list</i>.